

Research Proposal

Evidence-Gated Medical LLM Alerts for Pharmacogenomic Claims

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Thesis: This project studies one clinically interpretable workflow: LLM-drafted pharmacogenomic or genomic medication-alert text. The evaluation asks whether a pre-display evidence gate can reduce overclaiming relative to guideline-supported evidence without inappropriately denying bounded, aligned alerts.

Problem Statement

Medical LLMs can sound authoritative even when the evidence only supports a narrower statement. In pharmacogenomics this matters because a real guideline, a verified citation, a population-frequency result, or a variant assertion may support a medication-safety note without proving broad patient-outcome benefit, universal population transport, or deterministic action.

Hypothesis

Compared with ungated LLM alert text, a three-gate controller should reduce evidence overclaiming while tracking inappropriate denial in a future independent evaluation. The supported Stage 1 claim is narrower: the implementation follows its intended action grammar on author-designed synthetic archetypes, while also exposing two primary-gate precedence mismatches.

Methods

- Workflow: pharmacogenomic/genomic medication-alert text, including medication-gene pairs and variant assertions relevant to prescribing or safety review.
- Gates: citation/guideline support, population fit, and endpoint/actionability.
- Stage 1: 33 author-designed synthetic cases with expected decisions; executable controller reports action conformance, primary-gate conformance, designed overclaim archetypes blocked, bounded alerts allowed, inappropriate denial, one-gate ablation, and gate-precedence error categories. Stage 1 consumes structured evidence annotations only; free-text extraction is out of scope.
- Stage 2: replace author-designed cases with independently authored cases and blinded reviewer adjudication by qualified clinicians or pharmacogenomics reviewers.

Stage 1 Progress

Metric	Current result	Interpretation
Cases	33	Synthetic case bank for the pruned workflow, including three isolating gate cases.
Author-designed overclaim share	23/33	Stress-test coverage, not a measured clinical base rate.
Overclaim archetypes allowed unchanged	0/23	Derived from overclaim cases that reached allow unchanged.
Action / primary-gate conformance	33/33; 31/33	PGX19 and PGX24 are gate-precedence mismatches.
Inappropriate denial	0	Count of bounded aligned alerts blocked too strongly.

One-gate ablation	2, 1, 6	Overclaims allowed when disabling citation, population, or endpoint/actionability gates.
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Value

The contribution is valuable because pharmacogenomic alerts sit at the intersection of real guideline support, population transport, citation validity, and high-stakes medication decisions. A small auditable gate can turn an abstract concern about LLM overclaiming into a concrete evaluation plan for one recurrent clinical workflow.

Background Papers and Standards

- Byrne, Domenico, and Moore (2024), Artificial intelligence for improved patient outcomes.
- Stegenga (2018), Medical Nihilism.
- FDA-NIH BEST Resource, endpoint and surrogate endpoint definitions.
- CPIC CYP2C19-clopidogrel 2022 update.
- CPIC DPYD-fluoropyrimidines guideline update.

Guideline and Data Resources

- ML4H 2026 Call for Participation and Writing Guidelines.
- CPIC/ClinPGx pharmacogenomic guideline pages.
- ClinVar and gnomAD for variant/resource context.
- All of Us, Genome India, and Psifas/Mosaic as population-fit examples.